Online Resource for: The center and the ends of the elephant random walk with rare reversals

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Online Resource 1

This Online Resource accompanies "The center and the ends of the elephant random walk with rare reversals". Sections S1 to S3 give the second-order, tail and odd-N results. Section S4 proves the family estimates used for the mode bound, and Section S5 proves the second-difference results. These auxiliary proofs support the main paper and are part of the material to be refereed. Numbers prefixed S refer here; the other numbers refer to the paper. We use its notation throughout.

S1 The center and the crossing to second order

Theorem 2.6 gives $\log C_N$ up to an error $O(\varepsilon \log N)$. A lower bound that matches part (b) of that theorem brings the error down to $O(\varepsilon)$.

Proposition S1.1 (the center to second order)**.** *Let $N \geq 64$ be even. For every $0 < \varepsilon < \frac{1}{2}$,*

$$-11\varepsilon - \varepsilon^2(84 \log N + 1.05 \cdot 10^5) \leq \log C_N - \log \frac{4\varepsilon}{N+2} - 2\varepsilon \log N.$$

If moreover $\varepsilon(1+\log N) \leq \frac{1}{100}$, then the middle term is at most $5200\,\varepsilon$, and so $\left|\log C_N - \log \frac{4\varepsilon}{N+2} - 2\varepsilon \log N\right| \leq 5200\,\varepsilon$.

So C_N is $\frac{4\varepsilon}{N+2} N^{2\varepsilon}$ times a factor $1 + O(\varepsilon)$.

Idea of the proof. The proof is in Section S1.A. The upper bound is Theorem 2.6(b). For the lower bound we compare M after the first flip with Pólya's urn conditioned to end at h, which we call the bridge. Then $V_{t+1}(1)/\pi_t$ is the mean over the bridge of a product of factors, one for each step (Proposition S1.4). Each factor is 1 plus ε times a ratio that need not be bounded uniformly in N, and we control the factors through their first two moments under the bridge. By Jensen's inequality the mean of the product is at least the exponential of the mean of the sum of the logarithms. The step at time u adds about $2\varepsilon/u$ to this mean, and the sum over $u \leq N$ is about $2\varepsilon \log N$.

With this bound the proof of Theorem 2.7 gives one more term of the crossing.

Proposition S1.2 (the crossing to second order)**.** *As $N \to \infty$ over even N,*

$$x_N = g_N - \frac{2g_N \log N + g_N^2/2}{N(1+1/g_N)} + O\Big(\frac{\log N}{N}\Big).$$

More precisely, for every even N with $\log N \ge 30$ the error term is at most $20822 \log N/N$ in absolute value. So $x_N < g_N$ for every even N with $\log N \ge 3500$.

So the root g_N overshoots the crossing by an amount of order $\log^2 N/N$. The amount is $6 \log^2 N/N$ in the limit, but only about 60 percent of that at $N = 51200$ (Table 1).

Proof. We repeat the proof of Theorem 2.7(a) with one more term on each side. The center gains the term $2\varepsilon \log N$ and the end gains the term $-x^2/(2N)$, and both push F up.

Step 1. Both sides. Let F be as in the proof of Theorem 2.7, and let $1 \le x \le 2\Lambda_N$ and $\varepsilon = x/N$. For large N Proposition S1.1 applies, and its error $5200\,\varepsilon$ is $O(x/N)$, as is the term $\log(1+2/N)$ in $\log \frac{4\varepsilon}{N+2}$. For the end we use $\log(1-y) = -y - \frac{y^2}{2} + O(y^3)$ for $0 \le y \le \frac12$ with $y = x/N$. So

$$\log C_N = \log x + \log 4 - 2L + \frac{2xL}{N} + O\Big(\frac{x}{N}\Big), \qquad \log E_N = -\log 2 - x - \frac{x^2}{2N} + O\Big(\frac{x}{N}\Big).$$

In the second formula we multiplied the expansion by $N-1$, and the terms left over are $O(x/N)$ because x^2/N is bounded for $x \le 4L$. Subtracting gives

$$F(x) = G(x) - \Lambda_N + \frac{2xL + x^2/2}{N} + O\Big(\frac{x}{N}\Big) \qquad (1 \le x \le 2\Lambda_N).$$

Step 2. Solve for x_N. By Theorem 2.7, x_N lies in this range for large N. So $F(x_N) = 0$ gives $G(x_N) = \Lambda_N - \eta_N$ with $\eta_N = (2x_N L + x_N^2/2)/N + O(L/N)$. Since $G(g_N) = \Lambda_N$, the mean value theorem gives $x_N - g_N = -\eta_N/G'(\xi)$ for some ξ between x_N and g_N. Here $G'(\xi) = 1 + 1/\xi = (1+1/g_N)(1+O(1/N))$, since $|\xi - g_N| = O(L^2/N)$ and $g_N \ge L$ for large N. As $\eta_N = O(L^2/N)$, we get $x_N - g_N = -\eta_N/(1+1/g_N) + O(L^2/N^2)$. Replacing x_N by g_N in the main term of η_N changes it by $O(L^3/N^2)$, again because $|x_N - g_N| = O(L^2/N)$. All these errors are $O(L/N)$, which gives the formula. Its main term is of order L^2/N and is larger than the error for large N, so $x_N < g_N$.

Step 3. The explicit bound. Let $L \ge 30$. By the proof of Theorem 2.7, $24L^4 + 20800L^3 \le N$, and $d = x_N - g_N$ has $|d| \le \rho_N \le 1/L^2$. Let $1 \le x \le 4L$ and $\varepsilon = x/N$. Then $\varepsilon(1+\log N) \le \frac{1}{100}$, so Proposition S1.1 bounds $\log C_N - \log \frac{4\varepsilon}{N+2} - 2\varepsilon L$ by $5200\,\varepsilon$ in absolute value. Here $\log \frac{4\varepsilon}{N+2} = \log x + \log 4 - 2L - \log(1+\frac{2}{N})$ with $0 \le \log(1+\frac{2}{N}) \le \frac{2x}{N}$. For $0 \le y \le \frac12$ the series of the logarithm gives $|\log(1-y) + y + \frac{y^2}{2}| \le \frac{y^3}{3(1-y)} \le 2y^3$. So $(N-1)\log(1-\varepsilon) = -x + \varepsilon - \frac{x^2}{2N} + \frac{\varepsilon^2}{2} + R$ with $|R| \le 2N\varepsilon^3 = 2x^3/N^2 \le x/N$, because $2x^2 \le 32L^2 \le N$. Collecting terms, and using $\frac{\varepsilon^2}{2} \le \varepsilon$,

$$\Big|F(x) - G(x) + \Lambda_N - \frac{2xL + x^2/2}{N}\Big| \le 5200\varepsilon + \frac{2x}{N} + 2\varepsilon + \frac{x}{N} = \frac{5205\,x}{N}. \qquad (**)$$

Now $x_N \in [1, 4L]$ and $F(x_N) = 0$. Put $\eta = (2g_N L + g_N^2/2)/N$. Since $\Lambda_N/2 \le \Lambda_N - \log \Lambda_N \le g_N \le \Lambda_N$, we have $L/2 \le g_N \le 2L$, so $|d/g_N| \le \frac12$. Also $G(x_N) - \Lambda_N = d + \log(1 + d/g_N)$, and $d/g_N - (d/g_N)^2 \le \log(1+d/g_N) \le d/g_N$. The main term of $(**)$ at x_N differs from η by $d(2L + \frac{x_N + g_N}{2})/N$, which is at most $5L\rho_N/N \le L/N$ in absolute value. So $(**)$ at x_N gives

$$\Big|d\Big(1 + \frac{1}{g_N}\Big) + \eta\Big| \le \frac{5205\,x_N}{N} + \frac{L}{N} + \Big(\frac{d}{g_N}\Big)^2 \le \frac{20822\,L}{N},$$

since $5205\,x_N \le 20820L$ and $(d/g_N)^2 \le 4\rho_N^2/L^2 \le L/N$. The last inequality holds because $\rho_N \le 718L^2/N$ and $4\cdot 718^2 L \le 20800L^3 \le N$. Dividing by $1+1/g_N \ge 1$ gives $|x_N - g_N + \eta/(1+1/g_N)| \le 20822L/N$, which is the explicit bound.

Finally let $L \ge 3500$. The tangent of the logarithm at 8192 gives $\log\Lambda_N \le \Lambda_N/8192 + \log 8192 - 1$, so $g_N \ge \Lambda_N - \log\Lambda_N \ge 1.996L$. Then $2g_NL + g_N^2/2 \ge 5.984L^2$ and $1+1/g_N \le 1+1/(1.996L)$. As $20822L + 20822/1.996 < 5.984L^2$ for $L \ge 3500$, we get $\eta/(1+1/g_N) > 20822L/N$, and the explicit bound gives $x_N < g_N$. □

Remark S1.3 (numerical values and open points). The formula of Proposition S1.2 without its error term is within 0.07 of x_N already at $N = 400$ (Table S1). Proposition S1.1 does not identify the term of order ε. At fixed N the center is a polynomial $c_1\varepsilon + c_2\varepsilon^2 + \cdots$ in ε, by Proposition 2.5(b). From its Taylor coefficients, computed in double precision (`verify_elephant_crossing.py`), we get $c_2/c_1 - 2\log N = -1.585878$, -1.546572 and -1.541005 at $N = 1600$, 12800 and 51200. These values approach $2\gamma - 2 - \log 2 = -1.5387\ldots$, but we do not prove this limit here. We do not know whether, for some fixed k, the bound $\log C_N \le \log(c_1\varepsilon) + \varepsilon c_2/c_1 + O(\varepsilon^2(\log N)^k)$ holds uniformly in N in the range of Proposition S1.1. Table 1 has $x_N < g_N$ already at $N = 50$, while Proposition S1.2 proves it only for $\log N \ge 3500$.

Table S1: The crossings x_N of Table 1, their distance to $x_{2\mathrm{nd}}$, the formula of Proposition S1.2 without its error term, and the difference Υ_N of Theorem S1.7. These values are numerical and are not used in any proof.

N	x_N	$x_N - x_{2\mathrm{nd}}$	$\Upsilon_N \log N$
50	3.940487	+0.3452	−0.083
400	7.629217	+0.0621	−0.199
1600	10.240565	+0.0179	+0.215
3200	11.546834	+0.0097	+0.386
12800	14.158845	+0.0028	+0.607
51200	16.778468	+0.0008	+0.707

S1.A The bridge and the lower bound

Fix $1 \le t \le h$, and start the Pólya(c) chain at state 1 at time $t+1$. By Lemma A.1(b), given that it is at h at time N it moves up at time u from b with probability $\bar r_u(b) = \frac{h-b}{N-u}$, whatever c is. So its $h-1$ up steps are placed uniformly at random among the $N-t-1$ steps after time $t+1$. We call this law of the path $(M_u)_{t+1\le u\le N}$ the bridge, and we write $\mathbb{P}_t^{\mathrm{br}}$ and $\mathbb{E}_t^{\mathrm{br}}$ for it. For $1 \le b \le u-1$ let

$$\varpi_u^+(b) = 1 + \varepsilon\frac{u-2b}{b}, \qquad \varpi_u^-(b) = 1 - \varepsilon\frac{u-2b}{u-b},$$

$$\theta_u(b) = \frac{N(u-2b)^2}{2(N-u)\,b(u-b)}, \qquad Q_u(b) = (u-2b)^2\Big[\frac{\bar r_u(b)}{b^2} + \frac{1-\bar r_u(b)}{(u-b)^2}\Big],$$

and let $\varpi_u = \varpi_u^+(M_u)$ if $M_{u+1} = M_u + 1$ and $\varpi_u = \varpi_u^-(M_u)$ otherwise. Put

$$\mathcal{A}_t = \sum_{u=t+1}^{N-1} \mathbb{E}_t^{\mathrm{br}}\,\theta_u(M_u), \qquad \mathcal{B}_t = \sum_{u=t+1}^{N-1} \mathbb{E}_t^{\mathrm{br}}\,Q_u(M_u).$$

Both are nonnegative, and neither depends on ε.

Proposition S1.4 (bridge formula). *Let $0 \le \varepsilon \le 1$ and $1 \le t \le h$. Then*

$$V_{t+1}(1) = \pi_t \, \mathbb{E}_t^{\mathrm{br}}\Big[\prod_{u=t+1}^{N-1} \varpi_u \Big].$$

If $0 < \varepsilon \le \frac{1}{2}$, then $\log V_{t+1}(1) \ge \log \pi_t + \varepsilon \mathcal{A}_t - \varepsilon^2 \mathcal{B}_t$.

Proof. We divide V by the hitting probability of Pólya's urn. The quotient satisfies a recursion whose weights are the step probabilities of the bridge, and each step leaves one factor ϖ_u. For $u \ge t+1$ and $b \in \mathcal{S}_u$ we have $\Phi_u^0(b) > 0$, and we put $R_u(b) = V_u(b)/\Phi_u^0(b)$. By Lemma A.1(a) and (b) with $c = 0$, $\frac{b}{u}\Phi_{u+1}^0(b+1) = \bar r_u(b)\Phi_u^0(b)$ and $(1 - \frac{b}{u})\Phi_{u+1}^0(b) = (1 - \bar r_u(b))\Phi_u^0(b)$. Dividing $V_u = \mathcal{L}_u V_{u+1}$ by $\Phi_u^0(b)$ gives

$$R_u(b) = (1 - \bar r_u(b)) \, \frac{1 - r_u(b)}{1 - b/u} \, R_{u+1}(b) + \bar r_u(b) \, \frac{r_u(b)}{b/u} \, R_{u+1}(b+1),$$

where a term with an infeasible state is absent, and its bridge probability is 0 (if $b = h$ then $\bar r_u(b) = 0$, and if $u - b = h$ then $\bar r_u(b) = 1$). Also $\frac{r_u(b)}{b/u} = \varpi_u^+(b)$ and $\frac{1 - r_u(b)}{1 - b/u} = \varpi_u^-(b)$. So $R_u(b) = \mathbb{E}_t^{\mathrm{br}}[\varpi_u R_{u+1}(M_{u+1}) \mid M_u = b]$, and $R_N(h) = 1$. Iterating from $u = t+1$ and using $\Phi_{t+1}^0(1) = \pi_t$ gives the formula.

Now let $0 < \varepsilon \le \frac{1}{2}$. We take the logarithm of the formula and bound it from below in three steps. First, every factor is at least $\frac{1}{2}$. Indeed if $b \le u/2$ then $\varpi_u^+(b) \ge 1$ and $\varpi_u^-(b) - 1 \in [-\varepsilon, 0]$, and if $b \ge u/2$ the same holds with $+$ and $-$ exchanged. Second, by Jensen's inequality for the logarithm,

$$\log \mathbb{E}_t^{\mathrm{br}}\Big[\prod_u \varpi_u\Big] \ge \sum_u \mathbb{E}_t^{\mathrm{br}} \log \varpi_u \ge \sum_u \mathbb{E}_t^{\mathrm{br}}\big[(\varpi_u - 1) - (\varpi_u - 1)^2\big].$$

The second inequality is $\log(1+y) \ge y - y^2$ for $y \ge -\frac{1}{2}$, which holds since the difference vanishes at 0 and its derivative $y(1+2y)/(1+y)$ has the sign of y. Third, we compute the two means given $M_u = b$. Write $\bar r = \bar r_u(b)$. The bridge moves up with probability $\bar r$, and then $\varpi_u - 1 = \varepsilon \frac{u-2b}{b}$. Otherwise $\varpi_u - 1 = -\varepsilon \frac{u-2b}{u-b}$. Since $u\bar r - b = \frac{N(u-2b)}{2(N-u)}$, this gives

$$\mathbb{E}_t^{\mathrm{br}}[\varpi_u - 1 \mid M_u = b] = \varepsilon(u-2b)\frac{u\bar r - b}{b(u-b)} = \varepsilon\, \theta_u(b), \qquad \mathbb{E}_t^{\mathrm{br}}[(\varpi_u - 1)^2 \mid M_u = b] = \varepsilon^2 Q_u(b).$$

So the sum above is $\varepsilon \mathcal{A}_t - \varepsilon^2 \mathcal{B}_t$, which is the bound. □

Lemma S1.5 (the weights w_t). *For each fixed t, $w_t \to t2^{-t-1}$ as $N \to \infty$. For every even $N \ge 2$, $\sum_t w_t (t+1)^3 \le 124$.*

Proof. Each fraction in $\pi_t = \frac{t}{N-t}\prod_{i=1}^{t-1}\frac{h-i}{N-i}$ tends to $\frac{1}{2}$, so $N\pi_t \to t2^{1-t}$, and $Nc_1 \to 4$. This is the first claim. For the second claim we write w_t and $(t+1)^3$ through binomial coefficients in t. By the proof of Lemma 2.4 (with $v = t+1$ and $d = h$), π_t is a constant times $\binom{t}{1}\binom{N-1-t}{h-1}$, and the Chu–Vandermonde identity gives $\sum_t \binom{t}{j}\binom{N-1-t}{h-1} = \binom{N}{h+j}$. With $j = 1$ this gives the constant, namely $w_t = \binom{t}{1}\binom{N-1-t}{h-1}/\binom{N}{h+1}$. Now $t(t+1)^3 = 8\binom{t}{1} + 38\binom{t}{2} + 54\binom{t}{3} + 24\binom{t}{4}$, because both sides are polynomials of degree four that agree at $t = 0, 1, 2, 3, 4$. So $\sum_t w_t(t+1)^3$ is the sum of 8, 38, 54 and 24 times $\binom{N}{h+j}/\binom{N}{h+1}$ for $j = 1, 2, 3, 4$. Since $\binom{N}{h+j} \le \binom{N}{h+1}$ for $j \ge 1$, it is at most $8 + 38 + 54 + 24 = 124$. □

Under $\mathbb{P}_t^{\mathrm{br}}$ let $K = M_u - 1$ and $K' = m - K$ be the numbers of up steps and of other steps among the $m = u - t - 1$ steps between times $t+1$ and u. So K is hypergeometric, with m draws from $N - t - 1$ steps of which $h - 1$ are up steps. Let $\Omega_u = \{u/8 \le M_u \le 7u/8\}$.

Lemma S1.6. *Let $N \ge 16$, $1 \le t \le h$ and $t+1 \le u \le N-1$.*

(a) The second moment is

$$\mathbb{E}_t^{\mathrm{br}}(u - 2M_u)^2 = \frac{(t-1)^2(N-u)^2}{(N-t-1)^2} + \frac{4(u-t-1)(h-1)(h-t)(N-u)}{(N-t-1)^2(N-t-2)} \le (t-1)^2 + u.$$

(b) If $t \le \frac{N+2}{4}$, then $\mathcal{A}_t \ge 2(H_{N-1} - H_t - 1 - \frac{1}{t}) \ge 2(\log N - \log t - 3)$.

(c) If $t \le \frac{N}{8}$ and $u \ge 8(t+1)$, then $\mathbb{P}_t^{\mathrm{br}}(\Omega_u^c) \le 2e^{-0.0943u}$.

(d) If $t \le \frac{N}{8}$, then $\mathcal{B}_t \le 84(t + 1 + \log N) + 683(t+1)^3 + 19100$.

Proof. (a) Here $u - 2M_u = u - 2 - 2K$ and $u - 2 = m + t - 1$. Since $N - 2 = 2(h-1)$, the mean $\mathbb{E}K = \frac{m(h-1)}{N-t-1}$ gives

$$\mathbb{E}(u - 2M_u) = \frac{(m+t-1)(N-t-1) - m(N-2)}{N-t-1} = \frac{(t-1)(N-t-1-m)}{N-t-1} = \frac{(t-1)(N-u)}{N-t-1}.$$

The variance of a hypergeometric variable with m draws from $N-t-1$ steps, of which $h-1$ are up steps and $h-t$ are not, is $\operatorname{Var} K = \frac{m(h-1)(h-t)(N-t-1-m)}{(N-t-1)^2(N-t-2)}$, and $N - t - 1 - m = N - u$. So the second term is $4 \operatorname{Var} K$, and the identity is $\mathbb{E}(u-2M_u)^2 = (\mathbb{E}(u-2M_u))^2 + 4 \operatorname{Var} K$. For the bound, the first term is at most $(t-1)^2$ because $N - u \le N - t - 1$. In the second term $4(h-1)(h-t) = (N-2)(N-2t) = (N-t-1)^2 - (t-1)^2 \le (N-t-1)^2$, so it is at most $\frac{m(N-u)}{N-t-2}$. This is 0 for $u = t+1$, and for $u \ge t+2$ it is at most u, because $m \le u$ and $N - u \le N - t - 2$.

(b) We bound θ_u from below by a multiple of $(u-2b)^2$ and use only the variance term of (a). Since $b(u-b) \le u^2/4$, $\theta_u(b) \ge \frac{2N(u-2b)^2}{(N-u)u^2}$. By (a), $\mathbb{E}_t^{\mathrm{br}}(u-2M_u)^2 \ge 4 \operatorname{Var} K$, and in $4 \operatorname{Var} K$ the factor $N - u$ cancels. With $4(h-1)(h-t) = (N-2)(N-2t)$ this gives $\mathcal{A}_t \ge \nu_t \sum_{u=t+1}^{N-1} \frac{u-t-1}{u^2}$ with $\nu_t = \frac{2N(N-2)(N-2t)}{(N-t-1)^2(N-t-2)}$. We show that $\nu_t \ge 2$. Here $\nu_1 = \frac{2N}{N-3} > 2$, and for real $t \in [1, \frac{N+2}{4}]$, $\frac{d}{dt} \log \nu_t = -\frac{2}{N-2t} + \frac{2}{N-t-1} + \frac{1}{N-t-2} \ge \frac{3}{N-t-1} - \frac{2}{N-2t} \ge 0$, since the last inequality is equivalent to $4t \le N + 2$. Next, the sum is $H_{N-1} - H_t - (t+1)\sum_{u=t+1}^{N-1} u^{-2}$, which is at least $H_{N-1} - H_t - \frac{t+1}{t}$. The sum is also nonnegative. So multiplying by $\nu_t \ge 2$ gives the first inequality, whether this lower bound is positive or not. For the second inequality use $H_{N-1} - H_t \ge \log \frac{N}{t+1} \ge \log N - \log t - \frac{1}{t}$.

(c) Here $m \ge \frac{7}{8}u$, so $\frac{u}{8} \le \frac{m}{7}$. Since $t \ge 1$, $\mathbb{E}K \ge \frac{m}{2}$, and since $t \le \frac{N}{8}$, $\mathbb{E}K' = \frac{m(h-t)}{N-t-1} \ge \frac{3m}{8}$. Hoeffding's inequality also holds for sampling without replacement [1, Section 6], so $\mathbb{P}(K - \mathbb{E}K \le -\sigma m) \le e^{-2\sigma^2 m}$ for $\sigma > 0$, and the same holds for K'. Now $\{M_u < \frac{u}{8}\} \subset \{K - \mathbb{E}K < -\frac{5m}{14}\}$, and since $u - M_u = t + K'$, also $\{u - M_u < \frac{u}{8}\} \subset \{K' - \mathbb{E}K' < -\frac{13m}{56}\}$. So $\mathbb{P}(\Omega_u^c) \le e^{-2(5/14)^2 m} + e^{-2(13/56)^2 m} \le 2e^{-0.10778m} \le 2e^{-0.0943u}$.

(d) We split each term $\mathbb{E}_t^{\mathrm{br}} Q_u(M_u)$ of $\mathcal{B}_t$ into the part on Ω_u and the part off Ω_u, and the three terms of the bound come from the parts on Ω_u, the parts off Ω_u with $u < 8(t+1)$, and the parts off Ω_u with $u \ge 8(t+1)$. In all of them we use that $\bar{r}(u-b)^2 + (1-\bar{r})b^2 \le u^2$, so that $Q_u(b) \le \frac{(u-2b)^2 u^2}{b^2(u-b)^2}$. On Ω_u we have $b(u-b) \ge \frac{7u^2}{64}$, so $Q_u \le \frac{4096}{49}\frac{(u-2b)^2}{u^2} \le 83.6\frac{(u-2b)^2}{u^2}$. By (a) these parts add up to at most $83.6\sum_{u>t}(\frac{(t-1)^2}{u^2} + \frac{1}{u}) \le 83.6(t + H_{N-1}) \le 84(t+1+\log N)$. Off Ω_u we only have $b(u-b) \ge u - 1 \ge \frac{u}{2}$,

which gives $Q_u \le 4(u-2b)^2 \le 4u^2$. For $u < 8(t+1)$ we bound the probability of Ω_u^c by 1, and these parts add up to at most $\sum_{u<8(t+1)} 4u^2 \le \frac{4}{3}(8(t+1))^3 \le 683(t+1)^3$. For $u \ge 8(t+1)$, (c) bounds the part by $8u^2 e^{-0.0943u}$, and with $q = e^{-0.0943}$ we have $8\sum_{u\ge1} u^2 q^u = 8q(1+q)/(1-q)^3 < 19100$. □

Proof of Proposition S1.1. The upper bound is Theorem 2.6(b). For the lower bound let $0 < \varepsilon < \frac{1}{2}$ and $t_0 = \lfloor N/8 \rfloor \ge 8$. Every step below uses only this range of ε. By Proposition 2.5(a), $C_N = \varepsilon c_1 \sum_t w_t (1-\varepsilon)^{t-1} V_{t+1}(1)/\pi_t$. This is εc_1 times a mean over the probability vector (w_t). By Jensen's inequality the logarithm of the mean is at least the mean of the logarithms. For $t \le t_0$ we bound $\log(V_{t+1}(1)/\pi_t)$ from below by Proposition S1.4. For $t > t_0$ we use only that it is ≥ 0, which is Lemma A.1(d). This gives

$$\log C_N \ge \log(\varepsilon c_1) + \sum_t w_t (t-1)\log(1-\varepsilon) + \sum_{t\le t_0} w_t\big(\varepsilon \mathcal{A}_t - \varepsilon^2 \mathcal{B}_t\big).$$

The first sum. By $\log(1-\varepsilon) \ge -\varepsilon - \varepsilon^2$ and Proposition 2.5(c), it is at least $-(\varepsilon+\varepsilon^2)\sum_t w_t(t-1) \ge -2\varepsilon - 2\varepsilon^2$.

The terms with $\mathcal{A}_t$. Since $t_0 \le \frac{N+2}{4}$, Lemma S1.6(b) gives $\mathcal{A}_t \ge 2(\log N - \log t - 3)$ for $t \le t_0$. Let $w_\le = \sum_{t\le t_0} w_t$. By Jensen's inequality $\sum_t w_t \log t \le \log \sum_t w_t t \le \log 3$, so

$$\sum_{t\le t_0} w_t\, \varepsilon \mathcal{A}_t \ge 2\varepsilon(\log N - 3) w_\le - 2\varepsilon \log 3 \ge 2\varepsilon \log N - 2\varepsilon(1-w_\le)\log N - 6\varepsilon - 2\varepsilon\log 3.$$

By Proposition 2.5(c), $1 - w_\le \le 2\sum_{t>t_0} t 2^{-t} = 2(t_0+2)2^{-t_0}$, and $N < 8(t_0+1)$. As $(t_0+2)2^{-t_0}\log(8t_0+8)$ decreases for $t_0 \ge 8$, we get $2\varepsilon(1-w_\le)\log N \le 4 \cdot 10 \cdot 2^{-8}\log 72\,\varepsilon < 0.67\,\varepsilon$.

The terms with $\mathcal{B}_t$. Here $t \le t_0 \le \frac{N}{8}$, so Lemma S1.6(d) applies. With $\sum_t w_t(t+1) \le 4$ (Proposition 2.5(c)) and Lemma S1.5 it gives $\sum_{t\le t_0} w_t \mathcal{B}_t \le 84(3+1+\log N) + 683\cdot 124 + 19100 = 84\log N + 104128$.

Collecting terms, and using $c_1 = \frac{4}{N+2}$,

$$\log C_N - \log\frac{4\varepsilon}{N+2} - 2\varepsilon\log N \ge -(2+0.67+6+2\log 3)\,\varepsilon - \varepsilon^2(84\log N + 104130),$$

which gives the lower bound since $2+0.67+6+2\log 3 < 11$. For the last claim, the hypothesis on ε gives $\varepsilon^2(84\log N + 1.05\cdot 10^5) \le \varepsilon\,\frac{84\log N + 1.05\cdot 10^5}{100(1+\log N)}$. The fraction decreases in N, and its value at $N = 64$ is below 205. Since $11 + 205 < 5200$, the two-sided bound follows. □

S1.B Comparison with the persistent walk

We compare with the persistent walk of the introduction of the paper, in which each step repeats the previous one with probability p and switches with probability $q = 1-p$. Here p and q refer to that walk. Put $T = Nq$. Its end is $p^{N-1}/2$, its crossing is p_N, and $z_N = N(1-p_N)$. So $x = N\varepsilon$ corresponds to T, and the crossing x_N corresponds to z_N. We compare the two crossings through the crossing equation of [2, Proposition 3.8], which says that for every $N \ge 2$

$$z_N + \tfrac{1}{2}\log z_N = L + c_0 + \frac{1}{8z_N} + O(L^{-2}), \qquad c_0 = \tfrac{1}{2}\log\tfrac{\pi}{8}, \tag{S1}$$

where the error term is at most $\frac{1}{4L^2} + \frac{L^2+L}{N}$ in absolute value, and that $L/2 \le z_N \le L$. This is the only result of [2] that we use. Doubling (S1) without its last two terms and adding $\log 2$ gives $2z_N + \log(2z_N) = \Lambda_N + \log(2\pi)$, so $2z_N - \log(2\pi)$ nearly solves $G(x) = \Lambda_N$. What is left is of order $1/L$.

Theorem S1.7. *Let z_N be the crossing value of T for the persistent walk with N steps, as above. As $N \to \infty$ over even N,*

$$\Upsilon_N := x_N - \big(2z_N - \log(2\pi)\big) = \frac{c_*}{\log N} + O\Big(\frac{\log\log N}{(\log N)^2}\Big), \qquad c_* = \tfrac{1}{2}\log(2\pi) - \tfrac{1}{4} = 0.668939\ldots$$

Proof. Write $\ell = \log(2\pi)$ and $\bar{z} = 2z_N$. We compute $G(\bar{z} - \ell)$ in three steps and then compare it with $G(x_N)$.

Step 1. The size of z_N. By the bounds after (S1), the error term there is $O(L^{-2})$ and $L/2 \le z_N \le L$. So $\log z_N = \log L + O(1)$ and $\frac{1}{8z_N} = O(1)$, and (S1) gives $z_N = L - \frac{1}{2}\log L + O(1)$. Hence $\bar{z} = 2L - \log L + O(1)$, and

$$\frac{1}{8z_N} = \frac{1}{8L} + \frac{L - z_N}{8Lz_N} = \frac{1}{8L} + O\Big(\frac{\log L}{L^2}\Big), \qquad \frac{1}{\bar{z}} = \frac{1}{2L} + O\Big(\frac{\log L}{L^2}\Big).$$

Step 2. The equation for $\bar{z}$. We double (S1) and add $\log 2$. The left side becomes $2z_N + \log z_N + \log 2 = \bar{z} + \log\bar{z}$. On the right, $2c_0 + \log 2 = \log\frac{\pi}{4}$, and $\frac{1}{4z_N} = \frac{1}{4L} + O(\log L/L^2)$ by Step 1. So $\bar{z} + \log\bar{z} = 2L + \log\frac{\pi}{4} + \frac{1}{4L} + O(\log L/L^2)$.

Step 3. The value $G(\bar{z} - \ell)$. We have $\log(\bar{z} - \ell) = \log\bar{z} + \log(1 - \ell/\bar{z})$, and $\log(1 - \ell/\bar{z}) = -\ell/(2L) + O(\log L/L^2)$ by Step 1. With Step 2 and $\log\frac{\pi}{4} - \ell = -\log 8$ this gives

$$G(\bar{z} - \ell) = \bar{z} + \log\bar{z} - \ell + \log\Big(1 - \frac{\ell}{\bar{z}}\Big) = \Lambda_N + \frac{1}{4L} - \frac{\ell}{2L} + O\Big(\frac{\log L}{L^2}\Big) = \Lambda_N - \frac{c_*}{L} + O\Big(\frac{\log L}{L^2}\Big).$$

Step 4. Compare with x_N. By Theorem 2.7(a), $G(x_N) = \Lambda_N + O(L^2/N)$. So $G(x_N) - G(\bar{z} - \ell) = c_*/L + O(\log L/L^2)$. Both x_N and $\bar{z} - \ell = 2z_N - \log(2\pi)$ are $2L + O(\log L)$, so $G'(x) = 1 + 1/x = 1 + O(1/L)$ between them. By the mean value theorem we may divide by G', and we get $x_N - (\bar{z} - \ell) = c_*/L + O(\log L/L^2)$. □

We computed z_N from the exact law of the persistent walk [2], with a tolerance of about 10^{-15}. The difference Υ_N changes sign between $N = 400$ and $N = 1600$, and $\Upsilon_N \log N$ is 0.707 at $N = 51200$ against the limit $c_* = 0.669$ (Table S1). So the asymptotic regime is reached slowly.

S2 The tail

We keep the conventions of Section 3 and Appendix B of the paper, and we use the lemmas of that appendix.

This section shows that the law falls like $x/(2m^2)$ beyond the humps.

Theorem S2.1 (the tail)**.** *Let N be even, and let $x = N\varepsilon \ge 1$ depend on N with $x\log^2 N/N \to 0$. Let $\omega_N \to \infty$ and $\delta'_N \to 0$. Then*

$$f_N(m) = \frac{x}{m(m+1)}\big(1 + o(1)\big) \qquad \text{and} \qquad P_N(N - m) = \frac{x}{2m^2}\big(1 + o(1)\big),$$

uniformly for $\omega_N x\log^2(x+1) \le m \le \delta'_N N$. We do not compute the absolute constants behind the $o(1)$ terms.

For m well beyond $x\log^2 x$ and well below N, the chance of landing in the bin at distance m from an end is about half the expected number of flipped families with m steps, namely $x/(2m^2)$. The half is there because the first step must point to that end.

Idea of the proof. The proof is below. For the lower bound we use the first flipped family, which has size m with probability about $x/(m(m+1))$ (Lemma B.3), while the other flips change the count only by a relative amount of order $x\log m/m$. For the upper bound we show $\mathbb{P}(M_N \ge m) \le \frac{x}{m}(1+O(x\log^2 m/m))$, and the unimodality of f_N turns this into a local bound. The mirror term $f_N(N-m)$ is at most $C_N = O(x/N^2)$, which gives the constant $\frac{1}{2}$.

Lemma S2.2 (late start)**.** *Let $j \ge 2$, $t \ge j-1$ and $z \ge 0$. Then $\mathbb{P}(M_t^{[j-1]} \ge z) \le \exp\{-\frac{j-1}{2}(\frac{z}{t} - \varepsilon \log \frac{2t}{j-1})\}$.*

Proof. This is a Chernoff bound with an exponent that changes with time. A chain that starts at 0 at the late time $j-1$ is like an urn that already holds $j-1$ balls of one color, so the fraction $M_k^{[j-1]}/k$ moves slowly. For this reason we take an exponent θ_k of size about $(j-1)/(2k)$.

Write $G_k = M_k^{[j-1]}$ and $\nu = (j-1)/2$. For $k \ge j-1$ let $\theta_k = \log\frac{k}{k-\nu}$, so that $e^{\theta_k} - 1 = \frac{\nu}{k-\nu}$. We claim that for $k \ge j-1$

$$\mathbb{E}e^{\theta_{k+1}G_{k+1}} \le e^{\varepsilon\nu/(k+1-\nu)}\,\mathbb{E}e^{\theta_k G_k}.$$

To see this, condition on $G_k = c$. The chain moves up with probability $r_k(c) \le \varepsilon + c/k$, and $1+y \le e^y$. So

$$\mathbb{E}\big[e^{\theta_{k+1}G_{k+1}} \,\big|\, G_k = c\big] = e^{\theta_{k+1}c}\big(1 + r_k(c)(e^{\theta_{k+1}} - 1)\big) \le \exp\Big\{c\Big(\theta_{k+1} + \frac{\nu}{k(k+1-\nu)}\Big) + \frac{\varepsilon\nu}{k+1-\nu}\Big\}.$$

It remains to check that $\theta_{k+1} + \frac{\nu}{k(k+1-\nu)} \le \theta_k$. Now $\theta_k - \theta_{k+1} = \log(1+w)$ with $w = \frac{\nu}{(k-\nu)(k+1)}$, and $\log(1+w) \ge \frac{w}{1+w} = \frac{\nu}{k(k+1-\nu)}$. This proves the claim.

We start from $G_{j-1} = 0$ and use the claim for $k = j-1, \dots, t-1$. Since $\sum_{k=j}^{t}\frac{1}{k-\nu} \le \log\frac{t-\nu}{\nu} \le \log\frac{2t}{j-1}$, we get $\mathbb{E}e^{\theta_t G_t} \le \exp\{\varepsilon\nu\log\frac{2t}{j-1}\}$. Finally $\theta_t = -\log(1-\frac{\nu}{t}) \ge \frac{\nu}{t}$, so Markov's inequality gives $\mathbb{P}(G_t \ge z) \le e^{-\nu z/t}\,\mathbb{E}e^{\theta_t G_t}$, which is the claim. □

Proposition S2.3 (cumulative upper bound)**.** *Assume (H). For $6 \le m \le N/2$ with $m \ge 16x\log m$ we have $\mathbb{P}(M_N \ge m) \le \frac{x}{m}\big(1 + 800\,\frac{x\log^2 m}{m}\big)$.*

Idea of the proof. Given the first flip at time j, the count M_N is at most $|\mathcal{T}_V| + Z_j$, the size of the first flipped family plus the count from the rest of the tree. The event $|\mathcal{T}_V| \ge m$ gives the main term x/m. The other terms come from a shift of $|\mathcal{T}_V|$ by Z_j, from a large Z_j, and from a late first flip, and each is smaller by a factor of order $x\log^2 m/m$.

Proof. Step 1. Split by the first flip. Let $J = \lfloor N/4\rfloor$ and $Z_j = Z_{N-1}^{(j)}$. First let $2 \le j \le J$. By Lemma 3.3, given $V = j$ the count M_N has the law of $|\mathcal{T}_V| - Y_{|\mathcal{T}_V|} + Z^{(j)}_{N-|\mathcal{T}_V|}$, where, given $V = j$, $|\mathcal{T}_V|$ has the law β_j and is independent of Y and $Z^{(j)}$. This is at most $|\mathcal{T}_V| + Z_j$, because $Y \ge 0$ and $Z^{(j)}$ is nondecreasing in time. So $M_N \ge m$ needs $Z_j > m/8$, or $Z_j \le m/8$ and $|\mathcal{T}_V| \ge m - Z_j$. Hence

$$\mathbb{P}(M_N \ge m \mid V = j) \le \mathbb{P}\Big(Z_j > \frac{m}{8}\Big) + \mathbb{E}\Big[\bar\beta_j(m - Z_j);\, Z_j \le \frac{m}{8}\Big].$$

We compare $\bar\beta_j(m-z)$ with $\bar\beta_j(m)$ for an integer $0 \le z \le m/8$. By (8), each step down from s to $s-1$ with $s \le m$ multiplies $\bar\beta_j$ by at most $\varrho_j := 1 + \frac{j-1}{N-m-j+2}$, and $\varrho_j \le 1 + \frac{4(j-1)}{N}$ because $N - m - j \ge N/4$. Since $\varrho^z - 1 \le z(\varrho-1)\varrho^z$ for $\varrho \ge 1$, we get $\bar\beta_j(m-z) \le \bar\beta_j(m)\varrho_j^z \le \bar\beta_j(m)\big(1 + z(\varrho_j - 1)\varrho_j^{m/8}\big)$. So the mean above is at most $\bar\beta_j(m)\big(1 + (\varrho_j - 1)\varrho_j^{m/8}\,\mathbb{E}Z_j\big)$.

On $\{V > J\}$ there is no mark on $2, \dots, J$, so by Lemma 3.1 M_N is a copy of $M_N^{[J]}$ independent of $\{V > J\}$. Altogether

$$\mathbb{P}(M_N \geq m) \leq \sum_{j=2}^{J} \mathbb{P}(V = j) \Big[\bar{\beta}_j(m) \big(1 + (\varrho_j - 1) \varrho_j^{m/8} \, \mathbb{E} Z_j\big) + \mathbb{P}\Big(Z_j > \frac{m}{8}\Big) \Big] + \mathbb{P}(V > J) \, \mathbb{P}\big(M_N^{[J]} \geq m\big).$$

This gives four terms, which we bound in turn. The main term is $\sum_j \mathbb{P}(V = j) \bar{\beta}_j(m) \leq \mathbb{P}(|\mathcal{T}_V| \geq m) \leq \frac{x}{m}$ by Lemma B.3.

Step 2. The shift term. This is the sum over $2 \leq j \leq J$ of $\mathbb{P}(V = j) \, \bar{\beta}_j(m) \, (\varrho_j - 1) \, \varrho_j^{m/8} \, \mathbb{E} Z_j$. We bound the five factors by

$$\mathbb{P}(V = j) \leq \varepsilon, \quad \bar{\beta}_j(m) \leq e^{-\frac{(j-1)(m-1)}{N-1}}, \quad \varrho_j - 1 \leq \frac{4(j-1)}{N}, \quad \varrho_j^{m/8} \leq e^{\frac{(j-1)m}{2N}}, \quad \mathbb{E} Z_j \leq x \log \frac{N}{j-1}.$$

The second is (7), the fourth follows from the third and $1 + y \leq e^y$, and the last is Lemma B.2. The two exponentials combine. Since $m \geq 6$ we have $\frac{m-1}{N-1} - \frac{m}{2N} \geq \frac{m-2}{2N} \geq \frac{m}{3N}$. So with $a = \frac{m}{3N}$ and $i = j - 1$ the shift term is at most

$$\frac{4\varepsilon x}{N} \sum_{1 \leq i < N/4} i \, e^{-ai} \log \frac{N}{i}.$$

Here $i \leq J - 1 < N/4$, so every term is nonnegative. In this sum only the first flips up to a time of order $1/a = 3N/m$ matter, because a later first flip rarely has a family of size m. To bound the sum we write $\log \frac{N}{i} = \log \frac{m}{3} + \log \frac{1}{ai}$. The first part is nonnegative, and $\sum_{i \geq 1} i e^{-ai} \leq a^{-2}$. The second part is positive only for $i < 1/a$. There $i \log \frac{1}{ai} \leq \frac{1}{ea}$, so these at most $1/a$ terms add up to at most $\frac{1}{ea^2}$. Hence the shift term is at most

$$\frac{4\varepsilon x}{N} \cdot \frac{\log m + 1}{a^2} = \frac{36 x^2 (\log m + 1)}{m^2}.$$

Step 3. The big-Z term. This is $\sum_{j=2}^{J} \mathbb{P}(V = j) \mathbb{P}(Z_j > m/8)$, and again $\mathbb{P}(V = j) \leq \varepsilon$. Recall that Z_j counts the disagreeing steps of a walk whose first $j - 1$ steps agree. For small j we only have the general tail bound of Lemma B.2, which is of order $x \log m / m$. For large j the late start makes $Z_j \geq m/8$ exponentially unlikely (Lemma S2.2). We split at $j_{\mathcal{K}} = \lfloor \mathcal{K} N / m \rfloor + 1$ with $\mathcal{K} = 32 \log(m/x)$. Note that $m \geq 16 x \log 6 > 28x$, so $\mathcal{K} \geq 2$.

Let $2 \leq j \leq j_{\mathcal{K}}$. Lemma B.2 at the integer $\lfloor m/8 \rfloor + 1$ gives $\mathbb{P}(Z_j > m/8) \leq 8x H_m / m$. There are at most $\mathcal{K} N / m$ such j, and $\mathcal{K} \leq 32 \log m$. So they contribute at most

$$\varepsilon \cdot \frac{\mathcal{K} N}{m} \cdot \frac{8 x H_m}{m} = \frac{8 \mathcal{K} x^2 H_m}{m^2}.$$

Let $j > j_{\mathcal{K}}$, so that $j - 1 > \mathcal{K} N / m$. We use Lemma S2.2 with $t = N - 1$ and the level $m/8$. Its exponent contains the term $\varepsilon \log \frac{2(N-1)}{j-1}$. Here $\frac{2(N-1)}{j-1} \leq \frac{2m}{\mathcal{K}} \leq m$, so this term is at most $\frac{x \log m}{N} \leq \frac{m}{16N}$, by the hypothesis $m \geq 16 x \log m$. The other term in the exponent is $\frac{m}{8(N-1)} \geq \frac{m}{8N}$. So the lemma gives $\mathbb{P}(Z_j \geq m/8) \leq e^{-(j-1)m/(32N)}$, which is a geometric sequence in j. Summing over $j - 1 \geq j_{\mathcal{K}} > \mathcal{K} N / m$, these j contribute at most

$$\varepsilon \, \frac{e^{-\mathcal{K}/32}}{1 - e^{-m/(32N)}} \leq \varepsilon \cdot \frac{x}{m} \Big(\frac{32N}{m} + 1 \Big) \leq \frac{x}{m} \cdot \frac{33x}{m}.$$

Here $e^{-\mathcal{K}/32} = x/m$ by the choice of $\mathcal{K}$, and we used $\frac{1}{1-e^{-y}} \le \frac{1}{y} + 1$ for $y > 0$ and $\varepsilon \le x/m$.

Step 4. No early flip. If there is no flip up to time J, the walk agrees with X_1 for about $N/4$ steps, and Lemma S2.2 makes $M_N \ge m$ exponentially unlikely in m. Namely $\mathbb{P}(V > J) = (1-\varepsilon)^{J-1} \le e^{\varepsilon - 3x/16}$, since $J \ge 3N/16$, and Lemma S2.2 with $j - 1 = J \in [\frac{3N}{16}, \frac{N}{4}]$ and $t = N$ gives $\mathbb{P}(M_N^{[J]} \ge m) \le \exp\{-\frac{3m}{32} + \frac{x}{8}\log\frac{32}{3}\}$. As $x \le m/28$ and $\varepsilon \le \frac{1}{4}$, the product of the two is at most $e^{1/4 - m/13}$.

Step 5. Collect. Put $\ell = \log m \ge \log 6 > 1.79$, so that $\ell^2 > 3.2$. We compare each of the four error terms with $x^2\ell^2/m^2$. First $36(\ell+1) \le 40\ell^2$. Next $\mathcal{K} \le 32\ell$ because $x \ge 1$, and $H_m \le 1 + \ell \le 1.6\,\ell$, so $8\mathcal{K}H_m \le 410\ell^2$. Also $33 \le 12\ell^2$. Finally $y^2 e^{-y} \le 4$ for $y \ge 0$, which with $y = m/13$ gives $e^{1/4-m/13} \le 676 e^{1/4}/m^2 \le 304\ell^2 x^2/m^2$, since $x \ge 1$ and $676e^{1/4} < 870 < 304 \cdot 3.2$. As $40 + 410 + 12 + 304 \le 800$, the four error terms add up to at most $800x^2\ell^2/m^2$, which is the claim. □

Proposition S2.4 (local upper bound)**.** *Let $N \ge 64$ and $x \ge 1$, and put $\bar{c} = \frac{1}{40}$ and $A_1 = 34000$. If $\varepsilon \log N \le \bar{c}$ and $A_1 x \log^2(x+1) \le m \le N/8$, then*

$$f_N(m) \le \frac{x}{m(m+1)}\big(1 + 20000\,\Delta(m)^{1/2}\big), \qquad \Delta(m) = \frac{x\log^2 m}{m} + \varepsilon \log N + \frac{m}{N}.$$

Idea of the proof. To the right of its mode f_N is nonincreasing, so $(k+1)f_N(m)$ is at most the probability of $\{m - k \le M_N \le m\}$, which is a difference of two tails. We bound the first tail from above by Proposition S2.3 and the second from below by summing Proposition B.4. Then we choose k to balance the errors.

Proof. Step 0. Three facts. Put $L_1 = \log(x+1) \ge \log 2$, so that $xL_1^2 \ge 0.48$, $m \ge 16200$ and $\log m \ge 9$. Also $\log N \ge 4$, so $\varepsilon \le \frac{1}{160}$. For $0 < A \le 59000$ we have $\log(AxL_1^2) \le 11 + 3L_1$, since $\log A \le 11$, $\log x \le L_1$ and $\log L_1 \le L_1$. First, $\log(x+2) \le 2L_1 \le 2.9L_1^2$, so $804x\log(x+2) \le 8A_1 xL_1^2 \le 8m \le N$, and Lemma B.5 gives $m_h \le 15400xL_1^2 \le m/2$. Second, let $m' \ge m/2 \ge s_0 := 17000xL_1^2$. As $\log s/s$ decreases for $s \ge e$, we get $16x\log m'/m' \le 16x(11+3L_1)/s_0 \le 1$, because $16(11+3L_1) \le 17000L_1^2$. So $m' \ge 16x\log m'$, and also $m' \ge 6$. Third, $\log^2 s/s$ decreases for $s \ge e^2$. At $s_1 = A_1 xL_1^2$ we have $\log^2 s_1 \le (11+3L_1)^2 \le 242 + 18L_1^2$, and $x(242 + 18L_1^2) \le s_1$. So $x\log^2 m/m \le 1$, and $\Delta(m) \le 1 + \frac{1}{40} + \frac{1}{8} \le 1.15$.

Step 1. A difference of tails. Let $0 \le k \le m/2$. Then $m - k \ge m/2 \ge m_h$, and Lemma B.1 with $a = m - k$ and $b = m$ gives $(k+1)f_N(m) \le \mathbb{P}(M_N \ge m - k) - \mathbb{P}(M_N \ge m + 1)$.

Step 2. Bound each tail. Write $\Delta = \Delta(m)$. Here (H) holds. Proposition S2.3 at $m - k$, which applies by Step 0, bounds the first term by

$$\frac{x}{m-k} + \frac{800x^2\log^2(m-k)}{(m-k)^2} \le \frac{x}{m-k} + \frac{3200x^2\log^2 m}{m^2} \le \frac{x}{m-k} + \frac{3200x\Delta}{m}.$$

For the second we sum Proposition B.4 over $m < \ell \le \lfloor N/4 \rfloor$. The main terms give $x\big(\frac{1}{m+1} - \frac{1}{\lfloor N/4\rfloor + 1}\big) \ge \frac{x}{m+1}\big(1 - \frac{8m}{N}\big)$. As $\frac{\log(\ell+1)}{\ell}$ decreases for $\ell \ge 3$ and $\sum_{\ell > m} \frac{1}{\ell(\ell+1)} \le \frac{1}{m+1}$, the terms with $\frac{x\log(\ell+1)}{\ell}$ cost at most $\frac{x}{m+1} \cdot 20x\frac{\log(m+2)}{m+1} \le \frac{x}{m+1} \cdot \frac{40x\log^2 m}{m}$. The terms with $\varepsilon \log N$ cost at most $\frac{x}{m+1} \cdot 20\varepsilon \log N$. The last terms are $\frac{20x}{\ell(\ell+1)}\ell e^{-\ell/9} \le \frac{20x}{m+1}e^{-\ell/9}$, and they cost at most $\frac{20x}{m+1} \cdot \frac{e^{-m/9}}{e^{1/9}-1} \le \frac{x}{m+1} \cdot 200e^{-m/9}$. Since $e^{-y} \le 2/y^2$ for $y > 0$ and $m \ge 16200$, we have $200e^{-m/9} \le 32400/m^2 \le 2/m \le 2x\log^2 m/m$. So $\mathbb{P}(M_N \ge m+1) \ge \frac{x}{m+1}(1 - 42\Delta)$. As $\frac{x}{m-k} - \frac{x}{m+1} = \frac{x(k+1)}{(m-k)(m+1)}$ and

$\frac{42x\Delta}{m+1} \le \frac{42x\Delta}{m}$, this gives

$$f_N(m) \le \frac{x}{(m-k)(m+1)} + \frac{3242\,x\Delta}{(k+1)m}.$$

Step 3. Choose k. Put $\delta = \Delta^{1/2}$ and $X = \frac{x}{m(m+1)}$. Note that $\frac{1}{m} \le \frac{x\log^2 m}{m} \le \Delta$, and $\frac{1}{m^2} \le \frac{2}{m(m+1)}$. If $\Delta \le \frac{1}{16}$, take $k = \lceil m\delta \rceil$, so that $k \le \frac{m}{4}+1 \le \frac{m}{2}$. Then $\frac{m}{m-k} \le 1+\frac{2k}{m} \le 1+2\delta+\frac{2}{m} \le 1+4\delta$, because $\frac{1}{m} \le \Delta \le \delta$. Also $\frac{\Delta}{k+1} \le \frac{\delta}{m}$. So $f_N(m) \le X(1+4\delta) + \frac{3242x\delta}{m^2} \le X(1+4\delta+6484\delta)$, which gives the claim. If $\Delta > \frac{1}{16}$, take $k = \lfloor m/2 \rfloor$, so that $m-k \ge \frac{m}{2}$ and $k+1 \ge \frac{m}{2}$. Then $f_N(m) \le 2X + \frac{6484x\Delta}{m^2} \le X(2+12968\Delta)$. Here $\frac{1}{4} < \delta \le 1.073$, so $1 < 4\delta$ and $\Delta \le 1.073\,\delta$. Hence $2+12968\Delta \le 1+4\delta+13915\delta$, which again gives the claim. □

Proof of Theorem S2.1. We check that Propositions B.4 and S2.4 apply on the whole range of m and that their error terms tend to 0 uniformly there. Then we show that the mirror term $f_N(N-m)$ is negligible.

Step 1. The hypotheses hold. Put $\ell_x = \log(x+1) \ge \log 2$ and let $\mathcal{R}_N$ be the range $\omega_N x\ell_x^2 \le m \le \delta'_N N$. Since $x\log^2 N/N \to 0$, for large N we have $\varepsilon(1+\log N) \le \min(\bar{c}, \frac{1}{100})$, $\omega_N \ge \max(A_1, 16)$, $\delta'_N \le \frac{1}{8}$ and $804x\log(x+2) \le N$. So (H) holds as well. If $\mathcal{R}_N$ is empty there is nothing to prove.

Step 2. The errors are small on $\mathcal{R}_N$. The function $\frac{x\log^2 m}{m}$ decreases for $m \ge e^2$. At $m = \omega_N x\ell_x^2$ we have $m \ge e^2$ and $\log m \le \log\omega_N + 3\ell_x$, as $\log x$ and $\log\ell_x$ are at most ℓ_x. So on $\mathcal{R}_N$, uniformly in $x \ge 1$, $\frac{x\log^2 m}{m} \le \frac{(\log\omega_N+3\ell_x)^2}{\omega_N\ell_x^2} \le \frac{1}{\omega_N}(\frac{\log\omega_N}{\log 2}+3)^2 \to 0$. Also $m \ge \omega_N \log^2 2 \to \infty$ on $\mathcal{R}_N$. Since $\log(m+1) \le 2\log^2 m$ for $m \ge 3$, Proposition B.4 has $e_1(m) \le C(\frac{x\log^2 m}{m} + \varepsilon\log N + me^{-m/9}) \to 0$ uniformly on $\mathcal{R}_N$. Also $\Delta(m) \le \frac{x\log^2 m}{m} + \varepsilon\log N + \delta'_N \to 0$ there, and Proposition S2.4 applies. So $f_N(m) = \frac{x}{m(m+1)}(1+o(1))$ uniformly on $\mathcal{R}_N$.

Step 3. The mirror term. Lemma B.5 gives $m_h \le A_0 x\ell_x^2 \le h$ for large N, since $x\ell_x^2 \le x\log^2(N+1) = o(N)$. As $N-m \ge h$, Lemma 3.4 gives $0 \le f_N(N-m) \le f_N(h) = C_N$. By Theorem 2.6(b) and $\varepsilon\log N \to 0$, $C_N \le \frac{4x}{N(N+2)}N^{2\varepsilon}e^{5200\varepsilon} = \frac{4x}{N^2}(1+o(1))$. On $\mathcal{R}_N$ this is at most $\frac{x}{m(m+1)}\cdot 8\delta'^2_N(1+o(1))$, since $m(m+1) \le 2\delta'^2_N N^2$. Hence $P_N(N-m) = \frac{1}{2}[f_N(m)+f_N(N-m)] = \frac{x}{2m(m+1)}(1+o(1))$ uniformly on $\mathcal{R}_N$, and $m(m+1) = m^2(1+o(1))$ because $m \to \infty$. □

S3 Odd N

We show that the crossing for odd N behaves as for even N. As in Proposition 3.27, $N \ge 3$ is odd, $c = 1-2\varepsilon$, $\widetilde{C}_N$ is the probability of each of the two central bins, and $x_N = N\varepsilon_N$ is the crossing.

Proposition S3.1 (odd N, the crossing). *(a) As $N \to \infty$ over odd N, $x_N + \log x_N = \Lambda_N + O(L^2/N)$. So both parts of Theorem 2.7 hold for odd N as well. More precisely, if $L \ge 30$, then $|x_N - g_N| \le \tilde{\rho}_N$ and $|x_N + \log x_N - \Lambda_N| \le \tilde{\rho}_N$, where $\tilde{\rho}_N = (24L^2+20812L)/N \le 1/L^2$. In particular the last claim of Corollary 3.11(b) holds at the crossing for every large odd N.*

(b) For every odd $N \ge 1689$ we have $x_N > 2$, and at the crossing $P_N(0) < P_N(1)$. So Corollary 3.22 applies to these N with $x = x_N$.

(c) There is N'_1 such that the law at the crossing is convex between the humps for every odd $N \ge N'_1$. We do not compute the threshold N'_1.

Proof. (a) We follow the proof of Theorem 2.7(a), with $\widetilde{F}(x) = \log\widetilde{C}_N - \log E_N$ at $\varepsilon = x/N$ in place of F. Let $1 \le x \le 2\Lambda_N$. For large N, Theorem 2.6 applies to the even number $N+1$ at this

ε and gives $\log C_{N+1} = \log \frac{4\varepsilon}{N+3} + O(\varepsilon L)$. By Proposition 3.27(a), $\log \widetilde{C}_N = \log C_{N+1} + O(1/N)$. Here $\log \frac{4\varepsilon}{N+3} = \log x + \log 4 - 2L + O(1/N)$, and $1/N \le xL/N$. So

$$\log \widetilde{C}_N = \log x + \log 4 - 2L + O\Big(\frac{xL}{N}\Big),$$

which is the estimate for $\log C_N$ in Step 1 of that proof. The formula $E_N = \frac{1}{2}(1-\varepsilon)^{N-1}$ holds for every N (Lemma 2.2). So $(*)$ holds for $\widetilde{F}$, and Step 2 of that proof gives $|x_N - g_N| = O(L^2/N)$ and the claim. Part (b) of Theorem 2.7 follows from part (a) as before.

For the explicit bound let $L \ge 30$. As in the proof of Theorem 2.7, $e^L \ge L^{12}/12!$ gives $26L^4 + 20802L^3 \le N$. Let $1 \le x \le 4L$ and $\varepsilon = x/N$. Since $\log(N+1) \le L+1$, we get $\varepsilon(1+\log(N+1)) \le 4L(L+2)/N \le \frac{1}{100}$, so Theorem 2.6 applies to $N+1$. By Proposition 3.27(a), $\log \widetilde{C}_N = \log C_{N+1} - \log(1-u)$ with $u = c/N \le \frac{1}{N}$, and $0 \le -\log(1-u) \le u + u^2 \le \frac{2}{N}$. Next $\log \frac{4\varepsilon}{N+3} = \log x + \log 4 - 2L - \log(1+\frac{3}{N})$ with $0 \le \log(1+\frac{3}{N}) \le \frac{3}{N}$. Theorem 2.6 gives $\log(1-2\varepsilon) \le \log C_{N+1} - \log \frac{4\varepsilon}{N+3} \le 2\varepsilon \log(N+1) + 5200\varepsilon$. Here $\log(1-2\varepsilon) \ge -2\varepsilon - 4\varepsilon^2$, $4\varepsilon^2 \le \frac{1}{N}$, $2\varepsilon \log(N+1) \le \frac{8L^2+8L}{N}$ and $5200\varepsilon \le \frac{20800L}{N}$. For the end, $\log E_N = (N-1)\log(1-\varepsilon) - \log 2$ lies between $-\log 2 - x - \frac{16L^2}{N}$ and $-\log 2 - x + \varepsilon$, by $-y - y^2 \le \log(1-y) \le -y$. Adding up, and using $\varepsilon \le \frac{4L}{N}$, we find that $\widetilde{F}(x) - G(x) + \Lambda_N$ lies between $-\frac{12L+4}{N}$ and $\frac{24L^2+20808L+2}{N}$. So $(*)$ holds for $\widetilde{F}$ with $\tilde{\rho}_N = (24L^2 + 20812L)/N$. Then $\tilde{\rho}_N \le 1/L^2$, both ranges in Step 2 of that proof are nonempty, and Step 2 gives $|x_N - g_N| \le \tilde{\rho}_N$ and $|G(x_N) - \Lambda_N| \le \tilde{\rho}_N$.

For the last claim, $x_N = 2\log N - \log\log N + O(1)$ for odd N, so $\log N \le x_N \le 2\log N$ for every large odd N, and Corollary 3.11(b) applies with $x = x_N$.

(b) We argue as in the proof of Proposition 3.7. Let $N \ge 1689$ be odd and take $\varepsilon = 2/N$. Then $\varepsilon(1+\log(N+1)) \le \frac{1}{100}$, so Theorem 2.6(b) for $N+1$ and Proposition 3.27(a) give

$$\widetilde{C}_N \le \frac{N}{N-1} C_{N+1} \le \frac{N}{N-1} \cdot \frac{8}{N(N+3)} (N+1)^{4/N} e^{10400/N}.$$

The right side decreases in N and is below $1.4 \cdot 10^{-3}$ at $N = 1689$. Also $E_N = \frac{1}{2}(1-\frac{2}{N})^{N-1} > 0.06$, as in that proof. So $\widetilde{C}_N < E_N$ at $\varepsilon = 2/N$, and hence $\varepsilon_N > 2/N$ and $x_N > 2 \ge 2(1-\varepsilon_N)$. Theorem 3.5 then gives $P_N(N-1) > E_N$ at the crossing, which is $P_N(0) < P_N(1)$ by symmetry. This is the hypothesis of the second case of Corollary 3.22.

(c) Theorem 3.23 holds for every large N, even or odd, and every ε with $\log N \le x \le 2\log N$. By (a), $\log N \le x_N \le 2\log N$ for every large odd N. So Theorem 3.23 applies at the crossing. □

S4 Proofs for the family estimates

These proofs establish the estimates stated in Appendix B of the paper. We retain its notation, including the pathwise construction of the chains.

Proof of Lemma B.1. By Lemma 3.4, $f_N(\ell) \ge f_N(b)$ for $a \le \ell \le b$ in the first case, and $f_N(\ell) \ge f_N(a)$ in the second. Sum over ℓ. □

Proof of Lemma B.2. We use the tree of Lemma 3.1 on $[t]$, with marks only on $v > k_0$. Every odd vertex lies in the family of a marked vertex, so $M_t^{[k_0]} \le \sum_{v \text{ marked}} |\mathcal{T}_v|$. We call a family large if its size is at least z, and small otherwise. If $M_t^{[k_0]} \ge z$, then some marked family is large, or the

small marked families have total size at least z. We use the union bound for the first event and Markov's inequality for the second. Marks and tree are independent, so

$$\mathbb{P}\big(M_t^{[k_0]} \ge z\big) \le \varepsilon \sum_v \mathbb{P}(|\mathcal{T}_v| \ge z) + \frac{\varepsilon}{z} \sum_v \mathbb{E}\big[|\mathcal{T}_v|; |\mathcal{T}_v| < z\big].$$

By the first sum in Lemma 2.4, with t in place of N, on average $\frac{t}{d(d+1)}$ vertices have a family of size d. So the first sum over v is $\sum_{d \ge z} \frac{t}{d(d+1)} \le \frac{t}{z}$, and the second is $\sum_{d=1}^{z-1} \frac{t}{d+1} = t(H_z - 1)$. Together they give $\frac{\varepsilon t H_z}{z}$. For the mean, only the vertices $v > k_0$ can be marked and $\mathbb{E}|\mathcal{T}_v| = t/v$, so $\mathbb{E}M_t^{[k_0]} \le \varepsilon \sum_{v=k_0+1}^{t} \frac{t}{v} \le \varepsilon t \log \frac{t}{k_0}$. □

Proof of Lemma B.3. By Lemma 3.3, $\mathbb{P}(|\mathcal{T}_V| = d) = \sum_j \varepsilon(1-\varepsilon)^{j-2}\beta_j(d)$, and $1 - (j-2)\varepsilon \le (1-\varepsilon)^{j-2} \le 1$. Apply the two sums of Lemma 2.4, and sum the upper bound over $d' \ge d$. □

Proof of Proposition B.4. We bound the inner sum of Lemma 3.3 from below for each early first flip, namely for $2 \le j \le J := \lfloor N/8 \rfloor$. The terms with $j > J$ are nonnegative and we drop them.

Step 1. One value of j. Fix $2 \le j \le J$ and put $\Gamma_d = d - Y_d + Z_{N-d}^{(j)}$ for $1 \le d \le N - j + 1$. The inner sum in Lemma 3.3 is $Q_j(m) = \mathbb{E}S_j$ with $S_j = \sum_d \beta_j(d)\mathbf{1}\{\Gamma_d = m\}$. Let $\mathcal{E}$ be the event $\{Z_{N-1}^{(j)} \le m-1,\ Y_{2m} \le m\}$. We claim that

$$S_j \ge \beta_j(m)\Big(\mathbf{1}_{\mathcal{E}} - \frac{2(j-2)}{N} Y_{2m}\Big).$$

Off $\mathcal{E}$ the right side is at most 0 and $S_j \ge 0$. So let $\mathcal{E}$ hold. As d grows by one, Y_d grows by 0 or 1 and $Z_{N-d}^{(j)}$ falls by 0 or 1. So every step of Γ lies in $\{-1, 0, 1\}$, and Γ cannot jump over m. It starts at $\Gamma_1 = 1 + Z_{N-1}^{(j)} \le m$ (note that $Y_1 = 0$). At $d^\circ = m + Y_{2m} \le 2m$ we have $\Gamma_{d^\circ} \ge d^\circ - Y_{d^\circ} \ge d^\circ - Y_{2m} = m$, since $Z^{(j)} \ge 0$ and Y is nondecreasing. (The index d° is allowed, because $2m + j \le N$.) So $\Gamma_d = m$ for some $d \le d^\circ$, and as β_j is nonincreasing, $S_j \ge \beta_j(d^\circ) = \beta_j(m + Y_{2m})$. It remains to compare $\beta_j(m+y)$ with $\beta_j(m)$ for $0 \le y \le m$. By (8),

$$\frac{\beta_j(m+y)}{\beta_j(m)} = \prod_{i=1}^{y}\Big(1 - \frac{j-2}{N-m-i}\Big) \ge 1 - \frac{2y(j-2)}{N}.$$

Here $N - m - i \ge N/2$, so each fraction is at most $\frac{2(j-2)}{N} \le \frac{1}{4}$, and a product of factors $1 - q_i$ with $0 \le q_i \le 1$ is at least $1 - \sum_i q_i$. This proves the claim. Now $\mathbf{1}_{\mathcal{E}} \ge 1 - \mathbf{1}\{Y_{2m} > m\} - \mathbf{1}\{Z_{N-1}^{(j)} \ge m\}$. Taking means in the claim gives

$$Q_j(m) \ge \beta_j(m)\big[1 - \mathbb{P}(Y_{2m} > m) - \mathbb{P}(Z_{N-1}^{(j)} \ge m)\big] - \frac{2(j-2)}{N}\,\mathbb{E}Y_{2m}\,\beta_j(m).$$

Step 2. The chance that Y or Z is large. Lemma B.2 gives

$$\mathbb{P}(Y_{2m} > m) \le 2\varepsilon H_{m+1}, \qquad \mathbb{P}(Z_{N-1}^{(j)} \ge m) \le \frac{xH_m}{m}, \qquad \mathbb{E}Y_{2m} \le 2\varepsilon m \log N.$$

None of these bounds depends on j. Put $p = 2\varepsilon H_{m+1} + xH_m/m$. If $p > 1$, then $e_1'(m) > 1$ and there is nothing to prove. So let $p \le 1$.

Step 3. Sum over j. By Lemma 3.3, $f_N(m) \ge \sum_{j=2}^{J} \mathbb{P}(V=j)\, Q_j(m)$. With Steps 1 and 2 this gives

$$f_N(m) \ge (1-p)\sum_{j=2}^{J} \mathbb{P}(V=j)\,\beta_j(m) - \frac{4\varepsilon m \log N}{N} \sum_{j=2}^{J} \mathbb{P}(V=j)\,(j-2)\,\beta_j(m).$$

In the second sum we use $\mathbb{P}(V=j) \le \varepsilon$ and the second sum in Lemma 2.4. So the second term is at most

$$\frac{4\varepsilon m \log N}{N} \cdot \frac{2x(N-m-1)}{m(m+1)(m+2)} \le \frac{x}{m(m+1)}\, 8\varepsilon \log N.$$

The first sum is $\mathbb{P}(|\mathcal{T}_V| = m)$ without the terms $j > J$. So by Lemma B.3 it is at least $\frac{x}{m(m+1)}(1 - \frac{2x}{m+2}) - \varepsilon \sum_{j>J} \beta_j(m)$.

Step 4. The late first flips $j > J$. A family that starts after time $N/8$ rarely reaches size m. By (7), $\beta_j(m) \le \frac{j-1}{N-m} r^{j-1}$ with $r = e^{-a}$ and $a = \frac{m-1}{N-1} \le \frac{1}{4}$. We sum this over $j-1 \ge J$. With $\sum_{i \ge J} i r^i \le r^J\big(\frac{J}{1-r} + \frac{1}{(1-r)^2}\big)$, $1 - e^{-a} \ge a/2$ and $N - m \ge \frac{3}{4}N$ we get $\varepsilon \sum_{j>J} \beta_j(m) \le \frac{4\varepsilon}{3N}\big(\frac{2J}{a} + \frac{4}{a^2}\big) e^{-aJ}$. Since $\lfloor N/8 \rfloor \ge \frac{N-1}{9}$ for $N \ge 64$, we have $aJ \ge \frac{m-1}{9}$. Also $J \le N/8$ and $a \ge \frac{m-1}{N}$. So

$$\varepsilon \sum_{j>J} \beta_j(m) \le x\Big(\frac{1}{3(m-1)} + \frac{16}{3(m-1)^2}\Big) e^{-(m-1)/9} \le \frac{x}{m(m+1)}\, 20 m e^{-m/9}.$$

The last step uses $m+1 \le 3(m-1)$ and $m+1 \le 3(m-1)^2$, which hold for $m \ge 2$, and $e^{1/9} \le \frac{9}{8}$, since $17 \cdot \frac{9}{8} \le 20$.

Step 5. Collect. For $0 \le p \le 1$ and $q \ge 0$ we have $(1-p)(1-q) \ge 1 - p - q$. So Steps 3 and 4 give

$$f_N(m) \ge \frac{x}{m(m+1)}\Big(1 - p - \frac{2x}{m+2} - 20 m e^{-m/9} - 8\varepsilon \log N\Big),$$

which is the bound with $e_1'(m)$. For $e_1' \le e_1$ we use $\log(m+1) \ge \log 3 > 1$ and $\log N \ge \log 64 > 4$. So $H_m \le 1 + \log m \le 2\log(m+1)$, $H_{m+1} \le 1 + \log N \le \frac{5}{4}\log N$ and $\frac{2x}{m+2} \le \frac{2x \log(m+1)}{m}$. Hence $e_1'(m) \le \frac{4x\log(m+1)}{m} + \frac{21}{2}\varepsilon \log N + 20 m e^{-m/9} \le e_1(m)$. □

Proof of Lemma B.5. We pick a point ℓ of order $x \log x$ at which $f_N(\ell)$ is at least half of $\frac{x}{\ell(\ell+1)}$. Up to its mode f_N is nondecreasing. So if the mode were far to the right of ℓ, the bins from ℓ to the mode would carry more mass than the tail $\mathbb{P}(M_N \ge \ell)$ has.

Put $L_2 = \log(x+2) \ge \log 3 > 1$ and $\ell = \lceil 200 x L_2 \rceil$, so that $200 \le \ell \le 201 x L_2$ and $4\ell \le 804 x L_2 \le N$. Since $\log N \ge 4$, we have $\varepsilon \le \frac{1}{160}$. We bound the five terms of $e_1'(\ell)$ in Proposition B.4. As $(1+\log s)/s$ decreases for $s \ge 1$, and $\log 200 \le 6$, $\log x \le L_2$ and $\log L_2 \le L_2$, we get $\frac{xH_\ell}{\ell} \le \frac{x(1+\log(200xL_2))}{200xL_2} \le \frac{7+2L_2}{200L_2} \le \frac{9}{200}$. Next $\frac{2x}{\ell+2} \le \frac{1}{100}$, $2\varepsilon H_{\ell+1} \le 2\varepsilon(1+\log N) \le \frac{1}{80} + \frac{1}{20}$ and $8\varepsilon \log N \le \frac{1}{5}$. Finally $s e^{-s/9}$ decreases for $s \ge 9$, so $20\ell e^{-\ell/9} \le 4000 e^{-200/9} < 10^{-5}$. The five terms add up to less than $\frac{1}{2}$, so $f_N(\ell) \ge \frac{x}{2\ell(\ell+1)}$. If $m_h \ge \ell$, Lemmas B.1 (with $a = \ell$, $b = m_h$) and B.2 give $(m_h - \ell + 1) f_N(\ell) \le \mathbb{P}(M_N \ge \ell) \le x H_\ell / \ell$. So $m_h \le (\ell+1)(1 + 2H_\ell)$ in every case. Here $\ell + 1 \le 202 x L_2$, because $x L_2 \ge 1$. Also $H_\ell \le 1 + \log(201 x L_2) \le 7 + 2L_2 \le 9L_2$, so $1 + 2H_\ell \le 19 L_2$. Hence $m_h \le 3838\, x L_2^2$. Since $(x+1)^2 \ge x + 2$, we have $L_2 \le 2\log(x+1)$, and so $m_h \le 15352\, x \log^2(x+1) \le A_0 x \log^2(x+1)$. The last claim holds because $804 x \log(x+2) \le 4 A_0 x \log(x+2)$. □

S5 Proofs for the second difference

We keep the notation of Section 3. This section proves the results of Section 3.7 on the signs of the second difference.

S5.A Proof of Lemma 3.18

We insert the numbers η_i into ζ one at a time. If $\eta_i = 0$, the sign pattern does not change. If $\eta_i > 0$, then $\zeta_i > 0$ or $\zeta_{i+1} > 0$, so η_i is placed next to an entry of its own sign, and the pattern does not change. The case $\eta_i < 0$ is the same. This proves the first claim. Deleting entries from a list replaces its sign pattern by a subsequence, and η is obtained from w by deleting the entries ζ_i. This proves the second claim. □

S5.B Proof of Lemma 3.19

Put $\xi_t = P_t(1)/P_t(0)$. By (3), $P_{t+1}(0) = (1-\varepsilon)P_t(0)$ and $P_{t+1}(1) = (1-\varepsilon-\frac{c}{t})P_t(1) + \varepsilon P_t(0)$. So

$$\xi_{t+1} - \xi_t = \frac{1}{1-\varepsilon}\Big(\varepsilon - \frac{c\xi_t}{t}\Big), \qquad \xi_2 = \frac{2\varepsilon}{1-\varepsilon}.$$

We claim that $\xi_t < \varepsilon t/c$ for $t \geq 2$. For $t = 2$ this holds because $c < 1-\varepsilon$. If it holds for t, then $0 < \xi_{t+1} - \xi_t \leq \frac{\varepsilon}{1-\varepsilon} < \frac{\varepsilon}{c}$, and it holds for $t+1$. So ξ_t is strictly increasing for $t \geq 2$, and $\xi_t < 1$ whenever $t \leq c/\varepsilon$.

Now (4) with $a = 1$, and $D_t(0) = P_t(1) - 2P_t(0)$, give for $t \geq 2$

$$s_{t+1} = \big(1 - r_t(2)\big)s_t + \sigma_t, \qquad \sigma_t = \Big(\varepsilon - \frac{c}{t}\Big)(\xi_t - 2)\, P_t(0).$$

Here $1 - r_t(2) \geq 1 - \varepsilon - c = \varepsilon > 0$. If $t < c/\varepsilon$, both factors of σ_t are negative and $\sigma_t > 0$. If $t > c/\varepsilon$, the first factor is positive, and the second is first negative and then positive, because ξ_t increases. So $\mathrm{sp}(\sigma_2, \sigma_3, \dots) \preceq \Pi_3$. Let $\Phi_t = \prod_{u=2}^{t-1}(1 - r_u(2)) > 0$. Then

$$\frac{s_t}{\Phi_t} = s_2 + \sum_{u=2}^{t-1} \frac{\sigma_u}{\Phi_{u+1}}, \qquad s_2 = 1 - 3\varepsilon.$$

If $\varepsilon < \frac{1}{3}$, the terms $s_2, \sigma_2, \sigma_3, \dots$ are first nonnegative, then nonpositive, then nonnegative. Their partial sums are nonnegative and nondecreasing in the first stretch, nonincreasing in the second, and nondecreasing in the third. So the sign pattern of the partial sums is a subsequence of Π_3. If $\varepsilon \geq \frac{1}{3}$, then $c/\varepsilon \leq 1$, so the terms are first nonpositive and then nonnegative, and the sign pattern of the partial sums is a subsequence of $(-,+)$. This proves the first claim.

For the second claim let $0 \leq \varepsilon \leq 1$. The proofs of (3) and (4) use only that step $t+1$ is $+1$ with probability $r_t(a)$ given $A_t = a$, so both hold for this ε. We compute $P_3(0) = \frac{1}{2}(1-\varepsilon)^2$ and $P_3(1) = P_3(2) = \frac{1}{2} - P_3(0)$, so $s_3 = P_3(0) - P_3(1) = (1-\varepsilon)^2 - \frac{1}{2}$. Suppose that $s_3 = 0$. Then $\varepsilon = 1 - 2^{-1/2}$ and $P_3(0) = P_3(1) = \frac{1}{4}$. By (4) with $t = 3$ and $a = 1$, $s_4 = r_3(-1)D_3(0) = (\varepsilon - \frac{c}{3})(P_3(1) - 2P_3(0)) = -\frac{5\varepsilon - 1}{12}$. This is not 0, because $\varepsilon \neq \frac{1}{5}$. (With P_t defined by (3), the same computation works for every real ε, because the other root $1 + 2^{-1/2}$ of s_3 is not $\frac{1}{5}$ either. We need only $0 < \varepsilon < \frac{1}{2}$.) □

S5.C Proof of Theorem 3.20

For $3 \le t \le N$ let $\mathcal{Z}_t$ be the list

$$\mathcal{Z}_t = \big(s_N, s_{N-1}, \dots, s_{t+1},\ s_t, D_t(2), D_t(3), \dots, D_t(t-2), s_t,\ s_{t+1}, \dots, s_{N-1}, s_N\big).$$

It has $2N-t-1$ entries. Its middle part is $(D_t(1), D_t(2), \dots, D_t(t-1))$, because $D_t(t-1) = D_t(1) = s_t$ by symmetry. The list $\mathcal{Z}_3$ is $(s_N, \dots, s_4, s_3, s_3, s_4, \dots, s_N)$, and $\mathcal{Z}_N$ is the list in the statement.

We check that $\mathcal{Z}_{t+1}$ is obtained from $\mathcal{Z}_t$ as in Lemma 3.18, for $3 \le t \le N-1$. The first $N-t$ entries of $\mathcal{Z}_{t+1}$ are $s_N, \dots, s_{t+1}$, and they are copies of the first $N-t$ entries of $\mathcal{Z}_t$. So $A_i = 1$ and $B_i = 0$ there. The next $t-2$ entries are $D_{t+1}(b)$ for $2 \le b \le t-1$. By (4),

$$D_{t+1}(b) = r_t(b-2)\, D_t(b-1) + \big(1 - r_t(b+1)\big) D_t(b),$$

and $D_t(b-1)$ and $D_t(b)$ are neighbors in the middle part of $\mathcal{Z}_t$, at the right places. The coefficients satisfy $r_t(b-2) \ge \varepsilon$ and $1 - r_t(b+1) \ge 1 - \varepsilon - c = \varepsilon$, since $2 \le b \le t-1$. The last $N-t$ entries of $\mathcal{Z}_{t+1}$ are $s_{t+1}, \dots, s_N$, and each is a copy of the entry of $\mathcal{Z}_t$ one place to its right. So $A_i = 0$ and $B_i = 1$ there.

By Lemma 3.18, $\mathrm{sp}(\mathcal{Z}_N) \preceq \mathrm{sp}(\mathcal{Z}_{N-1}) \preceq \cdots \preceq \mathrm{sp}(\mathcal{Z}_3)$. By Lemma 3.19 the sign pattern p of $(s_3, \dots, s_N)$ is a subsequence of Π_3, so it is one of the lists $()$, $(+)$, $(-)$, $(+,-)$, $(-,+)$ and $(+,-,+)$. The sign pattern of $\mathcal{Z}_3$ is p reversed, followed by p, with equal neighbors merged. In the six cases this is $()$, $(+)$, $(-)$, $(-,+,-)$, $(+,-,+)$ and $(+,-,+,-,+)$, and each is a subsequence of Π_5.

For f_N we argue in the same way with $D_t^f(m) = f_t(m+1) - 2f_t(m) + f_t(m-1)$ and the lists

$$\mathcal{Z}_t^f = \big(s_N^f, \dots, s_{t+1}^f,\ s_t^f, D_t^f(2), \dots, D_t^f(t-1),\ d_t, d_{t+1}, \dots, d_{N-1}\big), \qquad 2 \le t \le N,$$

where $s_t^f = D_t^f(1)$ and $d_u = (1 - r_u(u+1))D_u^f(u) = (\varepsilon - \frac{c}{u}) f_u(u-1)$. The entry $D_{t+1}^f(t)$ of $\mathcal{Z}_{t+1}^f$ equals $r_t(t-2)D_t^f(t-1) + d_t$ by (4), and the other entries are as before. The proof of Lemma 3.19 applies to s_t^f without change, with $\xi_t = f_t(1)/f_t(0)$, $\xi_2 = \frac{\varepsilon}{1-\varepsilon}$ and $s_2^f = 1 - 3\varepsilon$. The numbers d_u are negative for $u < c/\varepsilon$ and positive for $u > c/\varepsilon$. So $\mathrm{sp}(\mathcal{Z}_2^f)$ is a subsequence of $(+,-,+)$ followed by $(-,+)$, and $\mathcal{Z}_N^f = (D_N^f(1), \dots, D_N^f(N-1))$. As before $\mathrm{sp}(\mathcal{Z}_N^f) \preceq \mathrm{sp}(\mathcal{Z}_2^f)$, and a subsequence of $(+,-,+)$ followed by $(-,+)$ is a subsequence of Π_5 once equal neighbors are merged. This argument needs only $N \ge 2$, and for $N = 2$ the lists $\mathcal{Z}_2^f$ and $\mathcal{Z}_N^f$ are the same. □

S5.D Proof of Theorem 3.21

By symmetry $D_N(N - a_1) > 0 > D_N(N - a_0)$. Suppose that $D_N(a) \le 0$ for some a with $a_1 < a < N - a_1$.

The case $D_N(a) < 0$. Then the entries of $\mathcal{Z}_N$ at $a_0 < a_1 < a < N - a_1 < N - a_0$ have the signs $-,+,-,+,-$. So $(-,+,-,+,-) \preceq \mathrm{sp}(\mathcal{Z}_N) \preceq \Pi_5$ by Theorem 3.20. This is impossible, because a subsequence of Π_5 with five terms is Π_5 itself.

The case $D_N(a) = 0$. We use the lists $\mathcal{Z}_t$ of the proof of Theorem 3.20 and number the entries of $\mathcal{Z}_t$ from 1 to $2N-t-1$. So $D_t(b)$ stands at place $N - t + b$ for $1 \le b \le t-1$. For $3 \le t \le N-1$ let W_t be the list w of Lemma 3.18 for $\zeta = \mathcal{Z}_t$ and $\eta = \mathcal{Z}_{t+1}$. Then $\mathrm{sp}(W_t) = \mathrm{sp}(\mathcal{Z}_t) \preceq \Pi_5$. The entry of $\mathcal{Z}_{t+1}$ at place k is a combination of the entries of $\mathcal{Z}_t$ at the places k and $k+1$, with nonnegative coefficients. We call these two entries its parents. For $2 \le b \le t-1$ the parents of $D_{t+1}(b)$ are $D_t(b-1)$ and $D_t(b)$, and both coefficients are positive.

For $4 \le t \le N$ let B_t be the set of integers b with $\max(2, a-N+t) \le b \le \min(a, t-2)$. It is not empty, because $2 \le a \le N-2$. We prove by downward induction on t that there are places $p_1 < p_2 < p_4 < p_5$ in $\mathcal{Z}_t$ such that the entries there have the signs $-,+,+,-$, that $D_t(b) = 0$ for every $b \in B_t$, and that all the places $N - t + b$ with $b \in B_t$ lie strictly between p_2 and p_4. For $t = N$ we have $B_N = \{a\}$, and we take the places a_0, a_1, $N - a_1$ and $N - a_0$.

Let the claim hold for $t+1$, where $4 \le t \le N-1$, with the places $p'_1 < p'_2 < p'_4 < p'_5$. Let $b \in B_{t+1}$. Then $D_{t+1}(b) = 0$ is a combination of its two parents with positive coefficients. So the parents both vanish, or they have strictly opposite signs. In the second case one parent is negative. In W_t it stands next to $D_{t+1}(b)$, hence between the positive entries of $\mathcal{Z}_{t+1}$ at the places p'_2 and p'_4, and these stand between the negative entries at p'_1 and p'_5. So $(-,+,-,+,-) \preceq \mathrm{sp}(W_t) \preceq \Pi_5$, which is impossible. Hence $D_t(b-1) = D_t(b) = 0$ for every $b \in B_{t+1}$. Every element of B_t is of the form b or $b-1$ with $b \in B_{t+1}$, so D_t vanishes on B_t.

The entry of $\mathcal{Z}_{t+1}$ at p'_1 is negative, so one of its parents is negative. Let $p_1 \in \{p'_1, p'_1 + 1\}$ be its place in $\mathcal{Z}_t$, and choose p_2, p_4 and p_5 in the same way. Then $p_1 \le p'_1 + 1 \le p'_2 \le p_2$, and $p_1 \ne p_2$ because the signs differ. In the same way $p_4 < p_5$. Let q^- and q^+ be the smallest and the largest place $N - t - 1 + b$ with $b \in B_{t+1}$. The places $N - t + b$ with $b \in B_t$ lie between q^- and $q^+ + 1$. Since $p_2 \le p'_2 + 1 \le q^-$ and $p_4 \ge p'_4 \ge q^+ + 1$, and the entries at p_2 and p_4 are not zero, these places lie strictly between p_2 and p_4. This proves the claim for t.

The induction step with $t = 4$ shows that both parents of $D_5(b)$ vanish for every $b \in B_5 \subseteq \{2, 3\}$. The parents of $D_5(2)$ are s_4 and $D_4(2)$, and those of $D_5(3)$ are $D_4(2)$ and s_4. So $s_4 = D_4(2) = 0$. By (4) with $t = 3$ and $a = 2$, and since $D_3(1) = D_3(2) = s_3$, we have $D_4(2) = r_3(0)s_3 + \big(1 - r_3(3)\big)s_3 = 2\varepsilon s_3$. So $s_3 = 0$ as well, which contradicts Lemma 3.19. □

References

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